MILITARY SPECIFICATION

SERVO ASSEMBLY: 10545441

This Amendment forms a part of Military Specification MIL-S-50795(MU) dated 1 May 1973 and is approved for use by all Departments and Agencies of the Department of Defense.

Table I: Delete in its entirety and substitute new Table I.

TABLE I - INPUT POWER

Pin Number	Voltage
25, 27, 33, 41, 43	+15.0 Vdc ± 1%
32	-15.0 Vdc ± 1%
30	+24 Vdc ± 2%
13, 15, 18	+13 Vdc ± 1%
29, 31, 38	Ground
5	+20 Vdc ± 5%
22	+10 Vdc ± 2%
35	-10 Vdc ± 2%
3 and 4 (Ret)	115 Vac ± 5%, 400 Hz ± 1%
8, 9 (center tap), 21 (Ret)	48 Vac ± 5%, 400 Hz ± 1%, center tapped, in phase with 115 Vac

FSC 1285

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Figure 3: Delete in its entirety and substitute new Figure 3.

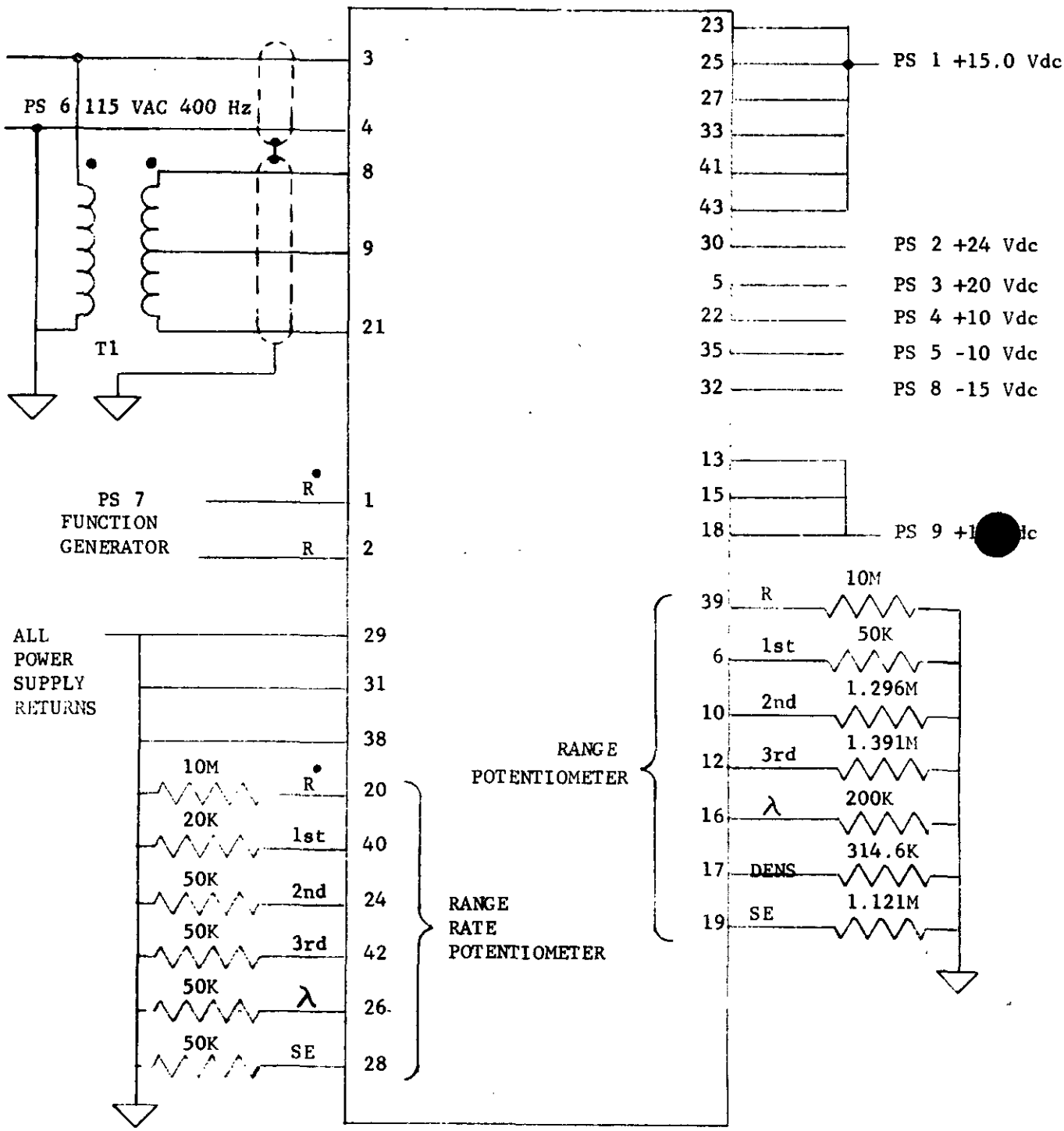


FIGURE 3 - TEST CONNECTIONS

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Table III: Delete in its entirety and substitute new Table III.

TABLE III - RANGE POTENTIOMETER R11

Cup	E In (Pin Number)	E Out (Pin Number)	Load
1 (Reference)	5	39	10 megohm ± 1%
2 (First Term)	32	6	50 kilohm ± .02%
3 (Second Term)	13	10	1.296 megohm ± .02%
4 (Third Term)	43	12	1.391 megohm
5 (Lambda)	15	16	200 kilohm
6 (Density)	41	17	314.6 kilohm
7 (Super Elev.)	18	19	1.121 megohm

NOTE: 1 all resistors .05%, ½ watt, unless otherwise specified.
2 the load shall be applied to pin designated E out.

Table IV: Delete in its entirety and substitute new Table IV.

TABLE IV - RANGE RATE POTENTIOMETER R12

Cup	E In (Pin Number)	E Out (Pin Number)	Load
1 (Reference)	22	20	10 megohm ± 1%
2 (Density)	23	40	20 kilohm ± .5%
3 (Second Term)	33	24	50 kilohm ± .1%
4 (Third Term)	33	42	50 kilohm ± .1%
5 (Lambda)	25	26	50 kilohm ± .1%
6 (Super Elev.)	27	28	50 kilohm ± .2%

NOTE: 1 all resistors are ½ watt.
2 load shall be applied to pin designated E out.

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Paragraph 3.2.1.2.1.1: Delete: "+1.6 Vdc" and "+19.40 Vdc" and substitute "+1.6 +0/-.2 Vdc" and "19.40 + .2/-0 Vdc."

Paragraph 3.2.1.2.1.2: Delete: "-9.7 Vdc" and "+9.7 Vdc" and substitute "-9.7 +0/-.2 Vdc" and "+9.7 +.2/-0 Vdc."

Paragraph 3.2.1.2.3.1: Add: "When the voltage level remains between +1.6 V and +19.4 V between pin 2 and pin 31."

Paragraph 3.2.1.3.2: Add: "When the voltage level remains between -9.7 V and +9.7 V between pin 1 and pin 31."

Paragraph 3.2.1.2.5: Delete in its entirety and substitute new paragraph 3.2.1.2.5.

"3.2.1.2.5 Range Potentiometer R11.- When power is applied per 3.2.1.1 and when measured according to Table III, the gains of the potentiometer cups shall be as specified in Table V, when calculated according to the following formula:

$$\text{Gain} = \frac{E_{out}}{E_{in}}$$

The tolerances of the gains are as specified in Table VII."

Paragraph 3.2.1.2.6: Delete in its entirety and substitute new paragraph 3.2.1.2.6.

"3.2.1.2.6 Range Rate Potentiometer R12.- When power is applied per 3.2.1.1 and when measured according to Table IV, the gains of the potentiometer cups shall be as specified in Table VI when calculated according to the following formula:

$$\text{Gain} = \frac{E_{out}}{E_{in}}$$

The tolerances of the gains are as specified in Table VIII."

Paragraph 3.2.1.3.3: Delete: "-40°F and +85°F" and substitute "+73°F and +85°F."

Add the following new paragraphs:

"4.2.3.2.13 Power Supply (PS-8).- The power supply shall be capable of producing -15 V ± .1% Vdc 100 milliamperes or greater."

"4.2.3.2.14 Power Supply (PS-9).- The power supply shall be capable of producing +13.0 ± .1% Vdc 100 milliamperes or greater."

Paragraph 4.2.5.1.1c: Delete: "pin 20 and pin 2" and substitute "pin 39 and pin 2."

Paragraph 4.2.5.1.1f: Delete: "pin 39 and pin 1" and substitute "pin 20 and pin 1."

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Paragraph 4.2.5.2: Delete in its entirety and substitute new paragraph 4.2.5.2:

"4.2.5.2 Test procedure.- With power applied per 3.2.1.1 and loads, inputs and outputs as defined in Table III, verify that the gains of the Range Potentiometer cups are within the tolerances in Table VII of the gains as specified in Table V. A minimum of three points, each cup, shall be verified."

Paragraph 4.2.5.3: Delete in its entirety and substitute new paragraph 4.2.5.3:

"4.2.5.3 Test procedure.- With power applied per 3.2.1.1 and loads, inputs and outputs as defined in Table IV, verify that the gains of the range rate potentiometer cups are within the tolerances in Table VIII of the gains as specified in Table VI. A minimum of three points, each cup, shall be verified."

Paragraph 4.3.3: Delete in its entirety and substitute new paragraph 4.3.3:

"4.3.3 Humidity.- The unit shall be tested for humidity performance per MIL STD 810, Method 507, Procedure 2."

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Add new Table VII as follows:

TABLE VII

(Approx) Angle	Ref Gain	Cup 2	Cup 3	Cup 4	Cup 5	Cup 6	Cup 7
0	.0667	.0025	.0185	.0120	.0150	.0150	.0100
30	.1467	.0025	.0175	.0120	.0150	.0150	.0100
60	.2267	.0028	.0170	.0050	.0150	.0120	.0100
90	.3067	.0028	.0165	.0050	.0150	.0120	.0100
120	.3867	.0030	.0160	.0050	.0150	.0090	.0100
150	.4667	.0020	.0150	.0040	.0100	.0080	.0100
180	.5467	.0020	.0100	.0040	.0100	.0060	.0100
210	.6267	.0010	.0050	.0050	.0100	.0060	.0100
240	.7067	.0010	.0050	.0060	.0090	.0050	.0100
270	.7867	.0010	.0050	.0060	.0090	.0050	.0100
300	.8667	.0010	.0050	.0060	.0090	.0050	.0100
330	.9467	.0010	.0050	.0060	.0090	.0050	.0100
350	1.000	.0010	.0050	.0060	.0090	.0050	.0100

NOTE: Tolerances are given with respect to the gain of the referenced cup, not the angle. The tolerance of the referenced cup is ± .0100 with respect to the shaft angle.

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Add new Table VIII as follows:

TABLE VIII

(Approx) Angle	Ref Gain	Cup 2	Cup 3	Cup 4	Cup 5	Cup 6
0	-1.000	.0075	.0025	.0100	.0080	.0100
30	-.8286	.0075	.0025	.0100	.0080	.0100
60	-.6571	.0075	.0025	.0100	.0080	.0100
90	-.4857	.0075	.0025	.0100	.0080	.0100
120	-.3143	.0075	.0025	.0100	.0080	.0100
150	-.1429	.0075	.0025	.0100	.0080	.0100
180	.0286	.0030	.0025	.0100	.0080	.0100
210	.2000	.0030	.0025	.0100	.0080	.0100
240	.3714	.0030	.0025	.0100	.0080	.0100
270	.5429	.0030	.0025	.0100	.0250	.0200
300	.7143	.0075	.0025	.0100	.0300	.0200
330	.8857	.0075	.0025	.0100	.0300	.0200
350	1.000	.0075	.0025	.0100	.0300	.0200

NOTE: Tolerances are given with respect to the gain of the referenced cup, not the shaft angle. The tolerance of the referenced cup is ± .0100 with respect to the shaft angle.

Custodian:

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Preparing Activity:

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Project No. 1285-A412

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